PRESTRESSED CONCRETE

Subject Code : 23A01504a

UNIT – II

LOSSES OF PRE-STRESS

Syllabus : Loss of Pre-Stress in Pre-Tensioned and Post-Tensioned Members due to Elastic Shortening, Shrinkage and Creep of Concrete, Relaxation of Stress in Steel, Anchorage Slip and Frictional Losses.

Contents

1. Introduction – types of losses
2. Elastic shortening of concrete
3. Shrinkage of concrete
4. Creep of concrete
5. Relaxation of steel
6. Anchorage slip
7. Frictional losses
8. Total losses – summary table (IS 1343)
9. Worked example – pre-tensioned beam
10. Worked example – post-tensioned beam
11. Practice problems & important questions

Exam tip : Learn the formula for each loss with the symbols, plus one numerical per loss. The full-loss numerical (10 marks) is guaranteed.

1. Introduction

The stress in the tendon immediately after jacking is not the same as the stress that remains in service. **The reduction of the initial prestressing force is called "loss of prestress".**

- Initial prestress P_i (at jacking) → P_0 (immediately after transfer) → P_e (effective, after all losses).
- Effective prestress $P_e = R \times P_i$ where **R = loss ratio / residual factor** (≈ 0.75 – 0.85).
- Losses are due to the shortening of concrete and the shortening / relaxation of steel; **losses can not be avoided** but can be estimated.
- Why estimate? – an **under-estimate** leads to cracking, an over-estimate leads to over-design.

$$\text{Loss \%} = (P_i - P_e)/P_i \times 100$$

Classification of losses

Pre-tensioned members	Post-tensioned members
1. Elastic shortening of concrete	1. Elastic shortening (partial – depends on sequence)
2. Shrinkage of concrete	2. Shrinkage of concrete
3. Creep of concrete	3. Creep of concrete
4. Relaxation of steel	4. Relaxation of steel
(no friction, no slip)	5. **Anchorage slip** 6. **Frictional loss** (length & curvature)

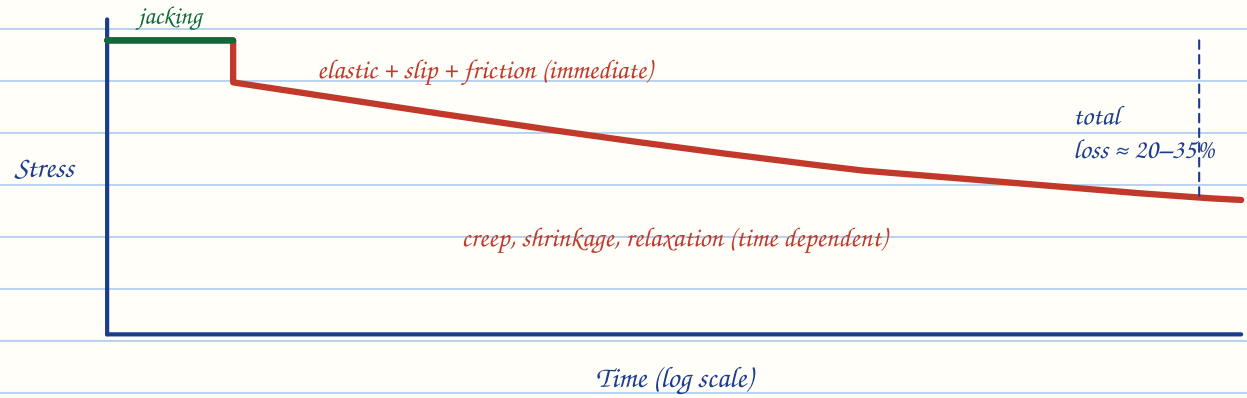


Fig. Variation of tendon stress with time (loss of prestress)

Immediate losses = elastic shortening, friction, anchorage slip. **Time-dependent losses** = creep, shrinkage, relaxation.

2. Elastic Shortening of Concrete

When prestress is transferred, concrete shortens elastically; the steel bonded to it (or anchored) shortens by the same strain → loss of stress in steel.

(a) Pre-tensioned members

- All tendons are released together – concrete shortens by the strain ε_c at the level of steel. The **steel stress drops by the same strain × E_s.**

$$\text{Loss} = \varepsilon_c E_s = (f_c/E_c) E_s = m f_c$$

- $m = E_s/E_c$ = modular ratio; $E_c = 5700\sqrt{f_{ck}}$ (IS 1343) N/mm²
- f_c = stress in concrete at the level of steel due to prestress (and self weight if effective):

$$f_c = P/A + P e^2/I \; (- M_g e/I)$$

where P = prestressing force after transfer ≈ P_i (approx.), A = area, I = M.I. of the section, e = eccentricity, M_g = moment due to self-weight (helps; reduces the stress at steel level – often neglected/taken).

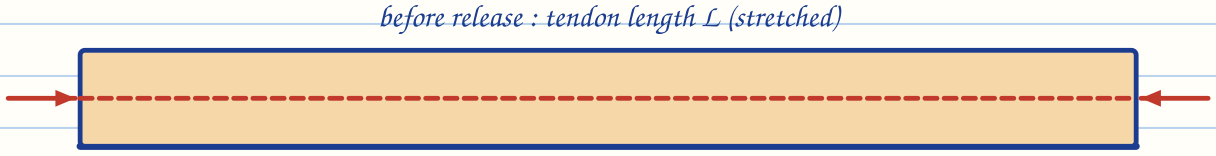


Fig. Elastic shortening – pre-tensioned member

Example 1

A pre-tensioned beam 100 mm × 250 mm is prestressed by 8 wires Ø 7 mm at a constant eccentricity of 50 mm, initial stress 1200 N/mm². $m = 6$. Find the loss due to elastic shortening.

$A_{ps} = 8 \times 38.5 = 308$ mm²; $P = 1200 \times 308 = 369.6$ kN; $A = 25000$ mm²; $I = 100 \times 250^3/12 = 1.302 \times 10^8$

$f_c = 369600/25000 + 369600 \times 50^2/1.302 \times 10^8 = 14.78 + 7.10 = 21.9$ N/mm²

Loss = $m f_c = 6 \times 21.9 =$ **131.3 N/mm²** (≈ 10.9 % of 1200)

(b) Post-tensioned members

- Tendons are stressed against hardened concrete – if **only one tendon** is stressed, the shortening occurs during jacking and is automatically compensated – **no loss**.
- If **several tendons** are stressed one after another, each subsequent tendon shortens the concrete and causes a loss in the previously stressed tendons. The **last tendon has no loss; the first one has the maximum**.

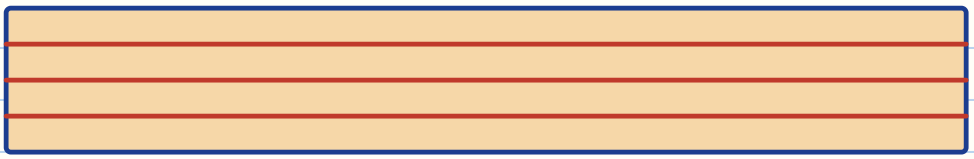


Fig. Sequential stressing of tendons

Let n = number of tendons, stressed successively. Stress in concrete at C.G. of steel due to all tendons = f_c:

- Loss in 1st tendon = $(n-1)/n \times m f_c$; last = 0.
- Average loss over all tendons (what we use):

$$\text{Loss} = \tfrac{1}{2} \cdot (n-1)/n \cdot m f_c$$

For $n = 1 \to 0$; for large $n \to \tfrac{1}{2}\, m f_c$. (Using $m f_c/2$ is a conservative approximation.)

Example 2

A post-tensioned beam, section 150 mm × 300 mm, has 4 cables stressed one after other, each with A_{ps} = 100 mm² at initial stress 1000 N/mm², constant e = 60 mm. $E_s = 2.1\times10^5$, $E_c = 3.5 \times 10^4$. Find the average loss.

$$P = 4 \times 100 \times 1000 = 400\ \text{kN};\ A = 45000;\ I = 150\times300^3/12 = 3.375\times10^8\ \text{mm}^4$$

$$f_c = 400000/45000 + 400000\times60^2/3.375\times10^8 = 8.89 + 4.27 = 13.16\ \text{N/mm}^2$$

$$m = 2.1\times10^5/3.5\times10^4 = 6.0;\ n = 4$$

$$\text{Loss} = \tfrac{1}{2}(3/4) \times 6 \times 13.16 = 29.6\ \text{N/mm}^2\ (\approx 3\ \%)$$

IS 1343 note : For pre-tension members elastic loss is based on the stress at transfer (steel level); for post-tension with several cables, loss may be calculated as above.

3. Shrinkage of Concrete

Shrinkage is the **contraction of concrete due to loss of moisture** (drying) and chemical changes – independent of stress. Steel shortens along with it → loss.

- Factors : w/c ratio, cement content & type, aggregate, humidity, size of member, age at transfer, curing. Higher w/c ratio → more shrinkage.
- **Occurs in both** pre- and post-tensioned members; the time-dependent shrinkage continues for years (~ 80 % in the first year).

IS 1343 values of residual shrinkage strain ε_{cs}

Type	Total shrinkage strain ε_{cs}
Pre-tensioning	0.0003
Post-tensioning	$0.0002 / \log_{10}(t + 2)$ (t = age at transfer, in days)

Loss due to shrinkage = $\varepsilon_{cs} \times E_s$

Post-tensioning loss is smaller because a part of shrinkage has already occurred before stressing (t = age of concrete at transfer). Older concrete → lower loss.

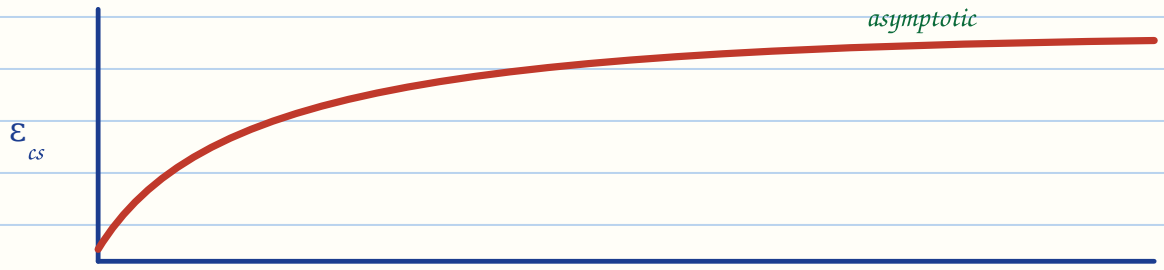


Fig. Shrinkage strain vs age – increases rapidly early, then flattens

Example 3

Calculate the shrinkage loss of (a) pre-tensioned member (b) post-tensioned member with age 10 days at transfer. $E_s = 2.1 \times 10^5$ N/mm².

(a) Pre : $0.0003 \times 2.1 \times 10^5 = 63$ N/mm²

(b) Post : $t = 10$; $\log_{10}(12) = 1.079$; $\varepsilon_{cs} = 0.0002/1.079 = 1.853 \times 10^{-4}$

Loss = $1.853 \times 10^{-4} \times 2.1 \times 10^5 = 38.9$ N/mm²

For the same steel, shrinkage loss in pre-tensioned = 63 N/mm² vs 30–40 N/mm² in post-tensioned (hence higher).

4. Creep of Concrete

Creep is the **time-dependent increase in strain under sustained stress**. Concrete under prestress continues to shorten with time → steel shortens → loss of prestress.

- Depends on : stress level (≈ proportional up to 0.4–0.5 f_{ck}), age at loading (earlier → more), humidity, w/c, aggregate, cement, size, duration.
- Creep strain $\varepsilon_{cc} = \theta \times (f_c / E_c)$ where θ = **creep coefficient** (ratio of creep strain to elastic strain).

Age at loading (transfer)	Creep coefficient θ (IS 1343)
7 days	2.2
28 days	1.6
1 year	1.1

$$\text{Loss due to creep} = \varepsilon_{cc} E_s = \theta \cdot (f_c / E_c) \cdot E_s = \theta \cdot m \cdot f_c$$

- f_c = stress in concrete at the level of centroid of steel (long-term sustained prestress – including self-weight effect).
- Applies to **both pre-tensioned and post-tensioned** members. Interpolate θ for intermediate ages.

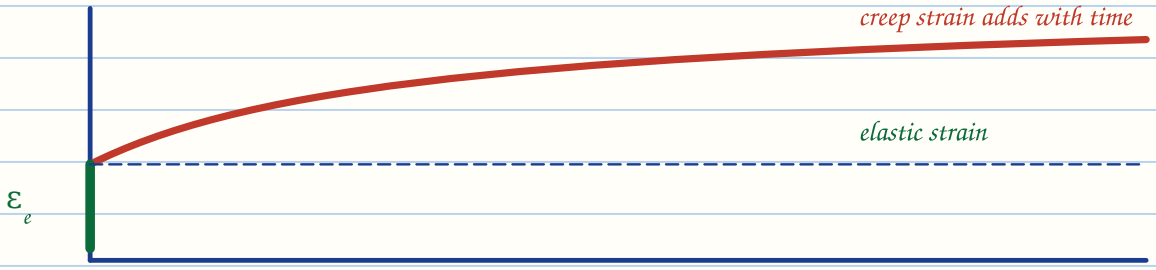


Fig. Creep strain under sustained stress

Example 4

A pre-tensioned beam has f_c = 7.5 N/mm² at the level of steel, transferred at 28 days, m = 6. Find creep loss.

$$\theta = 1.6 \text{ (28 days)} \rightarrow \text{Loss} = 1.6 \times 6 \times 7.5 = 72 \text{ N/mm}^2$$

If the same member is transferred at 7 days : 2.2 × 6 × 7.5 = 99 N/mm². (Earlier transfer = larger creep.)

Elastic, creep and shrinkage losses are all strain-based – they depend on E_s. That is why high initial stress (HTS) is needed.

5. Relaxation of Stress in Steel

Relaxation is the **decrease of stress with time in steel held at constant strain** (length). Prestressing tendons are held at constant length after anchoring – their stress gradually decreases.

- It is a **steel property** (like creep of steel), depending on initial stress, temperature, time and type of steel. It increases with initial stress and temperature.
- **Pure relaxation** (tested at constant length) is higher than **actual (reduced) relaxation** in structure because the steel length changes slightly due to creep / shrinkage; IS gives test values for 1000 h.
- Low-relaxation (stabilised) strands are used now: loss ≈ ¼ to ½ of the normal-relaxation steel.

IS 1343 relaxation losses (at 1000 hours)

Initial prestress	Relaxation loss (N/mm²)
$0.5\,f_{pk}$	0
$0.6\,f_{pk}$	35
$0.7\,f_{pk}$	70
$0.8\,f_{pk}$	90

(f_{pk} = characteristic tensile strength of the steel. Interpolate for intermediate values. Values are for normal-relaxation steel at 20 °C.)

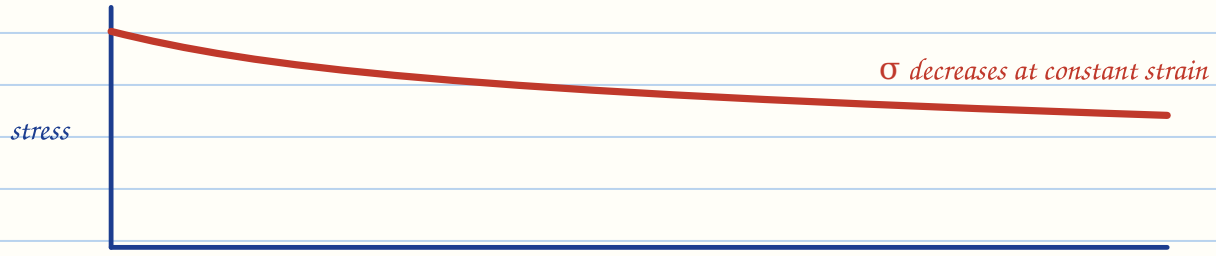


Fig. Relaxation of steel – stress vs time at constant length

Example 5

Initial stress in wires = 1200 N/mm², f_{pk} = 1600 N/mm². Find the relaxation loss.

$1200/1600 = 0.75\,f_{pk}$ → between 0.7 (70) and 0.8 (90)

Loss = 70 + (0.75 – 0.70)/(0.1) × 20 = **80 N/mm²** (6.7 % of 1200)

Relaxation loss ≈ 3 – 7 % of initial stress for normal steel; ≈ 1.5 – 2.5 % for low relaxation steel.

6. Anchorage Slip

Loss in **post-tensioned** members only. When the jack is released, the tendon draws a little into the anchorage as the wedges / cones seat themselves – a slip Δ (mm) – this reduces the tendon length (strain) → loss of stress.

- Typical slips (draw-in) : Freyssinet cone ≈ 4–6 mm; Gifford-Udall ≈ 1–2 mm; Magnel ≈ 1 mm; BBRV button head 0; threaded bars (Lee-McCall) 1 mm / nut.
- Slip influences a short length from the jacking end. Loss is greatest at the anchor and decreases towards the other end (friction also reverses).

Loss = (Δ / L) × E_s (for uniform strain; Δ in mm, L = length of tendon, mm)

- Δ = slip at anchorage (mm); L = length of the tendon (mm); E_s = modulus of steel.
- **Loss ∝ 1/L** – long tendons suffer less; short beams (small L) – larger %.
- Slip can be compensated by **over-tensioning** the jack by the extent of the expected slip.

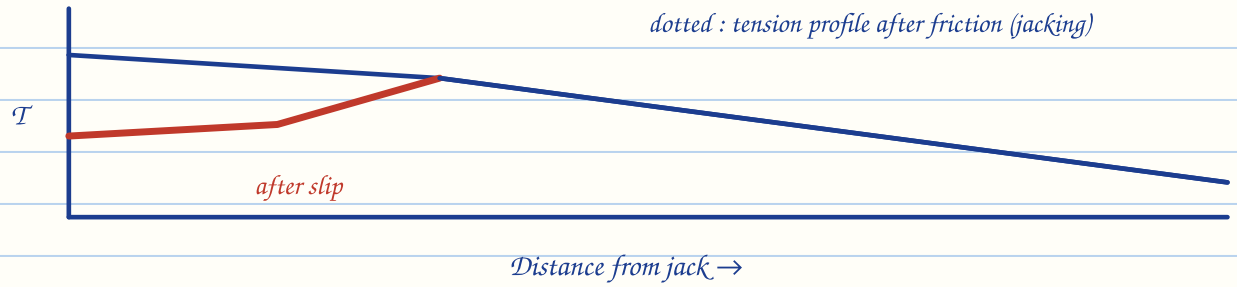


Fig. Effect of slip is confined to a length near the jack end

Example 6

A post-tensioned beam with tendons 20 m long is anchored with slip of 3 mm. $E_s = 2.1\times10^5$. Find loss of stress.

$$\text{Loss} = (3/20000) \times 2.1\times10^5 = 31.5\ N/mm^2$$

For L = 10 m – loss doubles (63 N/mm²). If initial stress = 1250 N/mm² → 2.5 % for 20 m.

7. Frictional Losses (post-tensioning only)

When tendon is stretched in a duct, friction between the tendon and duct reduces the force along its length. Force at the jack is largest, **smallest at the far (dead) end**.

Two components

1. **Length effect (wobble effect)** – because the duct can not be exactly straight; unintentional deviation. Depends on length x and wobble coefficient k.
2. **Curvature effect** – due to the intentional curvature of the duct; the tendon presses on the duct walls. Depends on the total angle turned α and coefficient of friction μ.

$$P_x = P_0 \cdot e^{-(\mu\alpha + kx)}$$

- P_0 = force at jack; P_x = force at distance x from jack; α = cumulative angle of curvature (radians) from jack to x;
- μ = coefficient of friction between tendon and duct; k = wobble coefficient per metre (also K).
- For small $(\mu\alpha + kx) \le 0.2$: $P_x \approx P_0 (1 - \mu\alpha - kx)$ (IS 1343 permits).

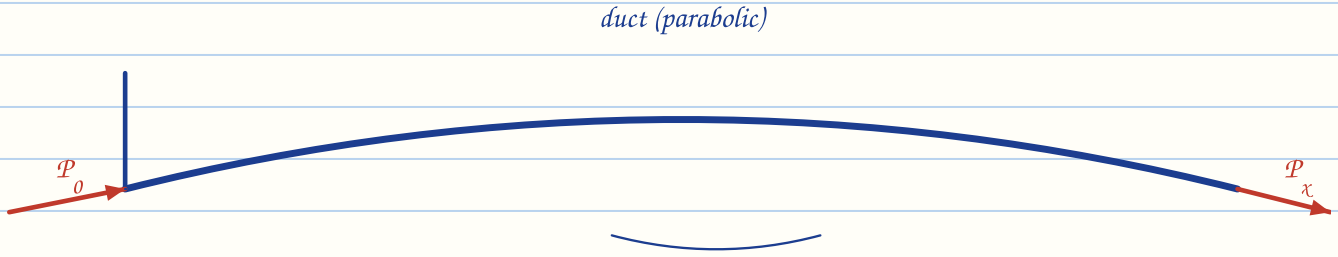


Fig. Tendon in a curved duct – friction reduces the force

IS 1343 coefficients

Type of duct / interface	μ
Steel moving on smooth concrete (formed with rubber tube)	0.55
Steel on steel (bright metal sheathing)	0.30
Steel on lead / greased	0.25

Wobble coefficient k = 0.0015 per metre (IS 1343, typical for sheathed ducts). Parabolic tendon : slope at end = $4e/L \rightarrow \alpha = 8e/L$ over whole span (for a symmetrical parabola with dip e).

Example 7 – Friction

A post-tensioned beam of span 20 m has a parabolic tendon with dip 0.4 m. P_0 = 1000 kN at the jack. μ = 0.35, k = 0.0015/m. Find the force and loss at the dead end.

$$\alpha = 8e/L = 8 \times 0.4/20 = 0.16 \text{ rad}$$

$$\mu\alpha + kx = 0.35 \times 0.16 + 0.0015 \times 20 = 0.056 + 0.030 = 0.086$$

$$P_x = 1000\, e^{-0.086} = 917.6 \text{ kN} \rightarrow \text{loss} = 82.4 \text{ kN } (8.24\ \%)$$

Approx. : 1000 (1 – 0.086) = 914 kN → loss 8.6 % (within accuracy).

Reducing friction : lubricate ducts, use bright metal sheathing, stress from both ends (loss at each end is half), use over-tensioning, mono-strands, use HDPE ducts, smaller curvature.

Stressing from both ends

Friction loss at the mid-span is lesser from both ends; the force profile is a "V" with equal values at both anchors. Thus the effective length for friction ≈ $L/2$ and loss halves. Commonly done for L > 25 m.

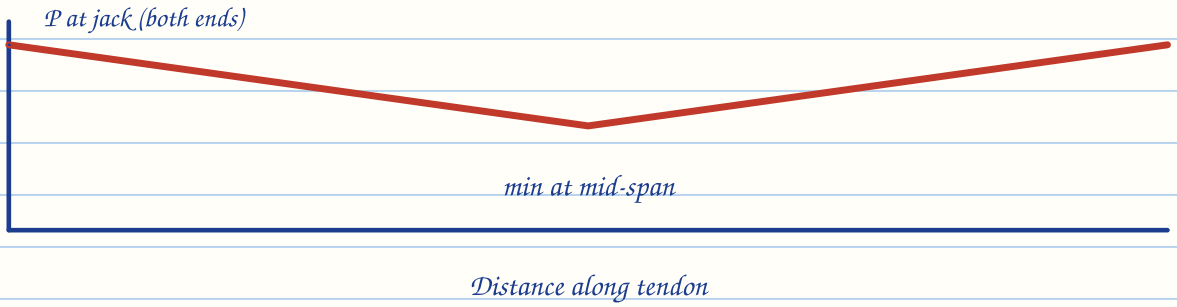


Fig. Tension profile when stressed from both ends

Frictional loss varies along length : 0 at jack, max at far end. Average loss ≈ half. Loss is automatically determined at design by the computed P_x at critical sections.

8. Total Losses – Summary

Type of loss	Pre-tensioned	Post-tensioned	Formula
Elastic shortening	Yes (full)	Partial (n cables)	$m f_c$; ½ $(n-1)/n\ m f_c$
Shrinkage	Yes (0.0003)	Yes (0.0002/log(t+2))	$\varepsilon_{cs} E_s$
Creep	Yes	Yes	$\theta\ m f_c$
Relaxation	Yes	Yes	IS table / % of f_{pi}
Anchorage slip	No	Yes	$(\Delta/L) E_s$
Friction	No	Yes	$P_0[1 - e^{-(\mu\alpha+kx)}]$

Typical percentage of total losses

Loss	Pre-tension	Post-tension
Elastic	3–5 %	1–3 %
Shrinkage	4–6 %	2–4 %
Creep	5–7 %	4–6 %
Relaxation	3–5 %	3–5 %
Slip	–	1–3 %
Friction	–	5–10 %
Total	**20–25 %**	**20–30 %**

IS 1343 (cl. 19.4) : If the losses are not calculated, a **lump-sum** may be assumed – generally **20 % for pre-tension** and **15 % for post-tension** (for concrete with relaxation low-level steel and no friction). This is only for preliminary design.

Effective prestress $P_e = P_i$ – (sum of losses) = $R \cdot P_i$

- **Time of loss :** elastic, friction, slip → at transfer (P_0). Creep, shrinkage, relaxation → over the years (P_e).
- Design checks : at transfer with P_0 (allowable stresses of concrete at transfer) and at service with $P_e = \eta\ P_0$.
- Typical **η = 0.80 – 0.85** (post-tensioned), 0.75 – 0.80 (pre-tensioned).

9. Worked Example – Pre-tensioned Beam (all losses)

A pre-tensioned beam 150 mm × 300 mm; 200 mm^2 of HTS wires at e = 50 mm. Initial stress = 1200 N/mm^2, f_{pk} = 1600 N/mm^2. f_{ck} = 40 N/mm^2, $E_s = 2.1 \times 10^5$. Transfer at 28 days. Estimate the total loss of prestress.

Step 1 – Section properties

$$A = 45000\ mm^2 ;\ I = 150 \times 300^3/12 = 3.375 \times 10^8\ mm^4$$

$$E_c = 5700\sqrt{40} = 36050\ N/mm^2 ;\ m = 2.1\times10^5/36050 = 5.83$$

$$P = 1200 \times 200 = 240\ kN$$

Step 2 – Stress in concrete at steel level

$$f_c = 240000/45000 + 240000 \times 50^2/3.375\times10^8 = 5.33 + 1.78 = 7.11\ N/mm^2$$

Step 3 – Losses

Loss	Working	N/mm²
Elastic shortening	$m f_c = 5.83 \times 7.11$	41.4
Shrinkage	$0.0003 \times 2.1\times10^5$	63.0
Creep (28 d, θ = 1.6)	$1.6 \times 5.83 \times 7.11$	66.3
Relaxation (0.75 f_{pk})	$70 + 0.5 \times 20$	80.0
Total		**250.7**

$$\%\ loss = 250.7/1200 \times 100 = 20.9\ \%$$

$$Effective\ prestress = 1200 - 250.7 = 949.3\ N/mm^2 \rightarrow P_e = 949.3 \times 200 = 189.9\ kN$$

Result matches the lump-sum 20 % rule for pre-tensioning. Residual factor η = 949.3/1200 = **0.79**.

10. Worked Example – Post-tensioned Beam

A post-tensioned beam 300 mm × 600 mm, span 20 m, parabolic tendon (ends 200 mm above the C.G., mid-span 200 mm below it, so dip = 0.40 m), A_{ps} = 800 mm², initial stress at jack 1250 N/mm², f_{pk} = 1770 N/mm². 4 cables stressed in sequence. f_{ck} = 50, $E_s = 2.1\times10^5$, μ = 0.35, k = 0.0015, slip = 3 mm, age at transfer = 28 d (shrinkage with t = 14 d). Find the total losses at the far end.

Section & material data

$$A = 1.8\times10^5 \text{ mm}^2;\ I = 300\times600^3/12 = 5.4\times10^9;\ E_c = 5700\sqrt{50} = 40305;\ m = 5.21$$

$$P = 1250\times800 = 1000 \text{ kN};\ f_c = 1000000/180000 + 10^6\times200^2/5.4\times10^9 = 5.56 + 7.41 = 12.96 \text{ N/mm}^2$$

Loss	Working	N/mm²
Elastic shortening	½ × (3/4) × 5.21 × 12.96	25.3
Shrinkage	$0.0002/\log_{10}(16)\times2.1\times10^5 = 1.661\times10^{-4}\times2.1\times10^5$	34.9
Creep (θ = 1.6)	1.6 × 5.21 × 12.96	108.1
Relaxation (0.706 f_{pk})	≈ 70 + 0.06 × 200 ≈ 70 (taken)	70.0
Anchorage slip	$3/20000\times2.1\times10^5$	31.5
Friction (α = 0.16, χ = 20)	$1250\,[1 - e^{-0.086}]$	103.0
Total		**372.8**

% loss = 372.8/1250 = 29.8 % (at far end). Effective stress = 877 N/mm²

Comments

- Friction and creep are the dominant losses in a long post-tensioned beam. The friction loss at the jack end is zero – the loss varies along the length; **design is done at critical section** (usually mid-span).
- At mid-span, friction ≈ 0.043 × 1250 ≈ 54 N/mm²; slip effect is zero if slip length < L/2 → total ≈ 290 N/mm² ≈ 23 %.
- Stressing from both ends would halve the friction loss and the slip effect of 31.5 N/mm² is retained at each end.

Always state clearly **where** (which section) the loss is being found. Note that creep depends on f_c at C.G. of steel.

11. Practice Problems & Important Questions

Numerical practice

1. A pre-tensioned beam 120 mm × 300 mm is prestressed by 4 wires Ø 7 mm at e = 60 mm, initial stress 1000 N/mm², m = 6. Find the loss due to elastic shortening. *(Ans ≈ 38 N/mm²)*
2. Estimate the shrinkage loss in (a) pre- (b) post-tensioned members of age 7 days for $E_s = 2\times10^5$. *(60 N/mm² ; 41.9 N/mm²)*
3. A concrete member transferred at 7 days has $f_c = 8$ N/mm² and m = 6. Find creep loss. *(2.2×6×8 = 105.6 N/mm²)*
4. A parabolic cable of 24 m span with dip 0.3 m and $P_0 = 800$ kN, $\mu = 0.3$, $k = 0.0015$. Find loss at the dead end. *($\alpha = 0.1; \Sigma = 0.066 \rightarrow 6.4$ %)*
5. A tendon of 15 m with slip 4 mm: find the loss. $E_s = 2\times10^5$. *(53.3 N/mm²)*

Short questions

1. What are the various types of losses in prestress ?
2. Why is there no friction loss in pre-tensioned members ?
3. Define relaxation, creep and shrinkage.
4. How is the loss due to elastic shortening reduced in post-tension ?
5. Mention the values of μ and k as per IS 1343.

Long questions

1. Explain in detail the various losses in pre-tensioned and post-tensioned members and the IS 1343 recommendations.
2. Derive the expression for friction loss $P_x = P_0 e^{-(\mu\alpha + kx)}$.
3. Compare the losses in pre- and post-tensioned members; mention how they can be minimised.
4. Numerical – compute all losses for a given beam (as in the worked example).

Formulae to memorise

Loss	Formula
Elastic (pre)	$m f_c$
Elastic (post)	$\frac{1}{2}\,(n-1)/n \cdot m f_c$
Shrinkage	$\varepsilon_{cs} E_s$; 0.0003 ; 0.0002/log(t+2)
Creep	$\theta\, m f_c$; θ = 2.2, 1.6, 1.1
Relaxation	IS 1343 table (0, 35, 70, 90)
Slip	$(\Delta/L) E_s$
Friction	$P_0\, e^{-(\mu\alpha + kx)}$